

SENTINEL

On-Premises Psychophysiological Triage Node

A hardware-software infrastructure that connects patients, psychologists, and trusted contacts through continuous biometric monitoring, AI-assisted journal analysis, and a deterministic crisis escalation protocol.

Independent Research Project | 2026

Companion documents: Sentinel Research Paper ([docs/sentinel_paper.pdf](#)), Sentinel Validation & Timing Report ([sentinel_validation_report.pdf](#)).

Technology White Paper

1. Executive Summary

Sentinel is an on-premises, hardware-software infrastructure for continuous mental health support. It unifies three stakeholders - the patient, the psychologist, and the trusted contact - into a single real-time ecosystem. A patient-facing portal captures biometric signals from wearable smart rings and AI-summarized journal entries. A clinician dashboard presents triage-ranked patients with AI-synthesized clinical notes and an active-crisis broadcast. A no-login trusted-contact page provides signed escalation links during crisis events.

The system was designed around one conviction: crisis support cannot depend on appointment availability. By fusing physiology, language, and self-report into one assessment pipeline, Sentinel surfaces cross-signal discrepancies (a patient writing "I'm okay" while biometric trends show escalating stress) and escalates deterministically through a staged crisis protocol. The platform is open-source, deploys to commodity hardware, runs its primary AI inference fully offline, and has a validated measurement burden of under 5 milliseconds for rule-based engines and 1.4-5.1 seconds for local LLM summarization.

2. The Problem

2.1 The Between-Sessions Gap

The therapeutic relationship is confined to weekly or biweekly sessions. Between appointments, patients experience emotional fluctuation and, in critical cases, crisis states without a channel for intervention. Telehealth platforms offer appointment-only connectivity; wellness apps provide generic content without clinician integration; crisis helplines operate as standalone services with no continuity of care.

2.2 Clinician Overload

WHO data indicates a global median of 0.75 psychiatrists per 100,000 people in low-resource settings. Clinicians carry caseloads that require continuous monitoring, documentation, triage, and follow-up - administrative burden that displaces therapeutic presence and drives burnout.

2.3 Fragmented Signals

Biometric wearables, journal text, and clinical notes live in separate silos. Cross-signal patterns - a physiological marker that precedes self-reported deterioration by 24-48 hours - remain undetected until they manifest as crisis.

3. The Sentinel Solution

Sentinel is a modular, dual-portal platform over a shared state layer (SQLite/PostgreSQL). Three integrated subsystems close every clinical loop inside the ecosystem.

3.1 Patient Portal

- Biometric trend visualization: heart rate, stress, sleep, SpO2, and mood.
- AI-summarized journaling with daily mood tracking and 28-label GoEmotions emotion classification.
- Session booking, follow-up tasks with proof-based completion, and a one-tap crisis trigger.

3.2 Psychologist Portal

- Triage-ranked patient list; crisis cases auto-expand with live state.
- Dual-mode AI documentation: warm patient reflection and structured OAP clinical notes.
- Follow-up grading (green/yellow/red), booking accept/waitlist queue, and an AI clinician sidebar (pre-session briefs, relapse indicators, silent-period watch, cross-patient patterns).

3.3 Trusted Contact Portal

- Standalone no-login page reachable only via HMAC-signed, expiring links.
- One-tap acknowledgment and "I'm on my way" response reflected live to the clinician.

3.4 Hardware Abstraction Layer

Sentinel does not build proprietary wearables. A pluggable RingSource SDK supports BLE GATT (bleak), vendor cloud APIs, and a deterministic simulator through one abstract base class. Devices authenticate via SHA-256 hashed device tokens compared in constant time. Hardware M0 (ring SDK + device binding) is complete; an OEM ring is in procurement.

4. Crisis Engine - Deterministic Escalation

The crisis engine implements a time-sensitive, multi-stage escalation protocol with no single point of failure: Stage 1 siren feedback on the patient device (0-29s); Stage 2 email to the trusted contact with a signed acknowledgment link (30s); Stage 3 helpline escalation (60s). Psychologist acknowledgment at any point halts all escalation, freezes the response timer, and records resolution duration.

Activation is driven by an explainable risk engine (score 1-10) blending keyword signals, emotion-classifier probabilities, and a temporal trend. Automatic activation fires at the crisis threshold and is throttled by a 3600-second cooldown so repeated entries cannot fire the full protocol repeatedly.

5. AI as Assistant, Not Replacement

Sentinel's design principle is that AI supports but never decides. The system does not diagnose, prescribe, or override clinical judgment. Functions are strictly supportive:

- Journal summarization in a warm patient mode (custom 7.2B-parameter therapy-tuned `sentinel` Ollama model).
- Clinical note synthesis into Observations-Assessment-Plan format.
- Local TF-IDF + LogisticRegression emotion classifier (GoEmotions, 28 labels, ~4 MB pickle) that feeds emotion hints into summarization.
- Echo detection that prevents the AI from parroting raw patient text (word-overlap >85% triggers a fallback).

- Three-tier fallback: local `sentinel` model, then Groq Cloud, then rule-based extraction - AI never fails silently.

5.1 Privacy-First Inference

The primary AI path runs entirely on local hardware via Ollama - no patient data leaves the local network, and no internet connection is required. Cloud AI is disabled by default and enabled only by explicit operator opt-in with encrypted transport.

6. Security and Data Protection

Sentinel treats clinical data protection as an architectural requirement, not an add-on:

- bcrypt password hashing with strict policy; per-account lockout after repeated failures.
- HS256 JWT access tokens with refresh-token rotation; HttpOnly cookie session storage.
- Field-level transparent encryption (Fernet) under a PBKDF2-600K-derived key activated by an operator passphrase ceremony; clinical notes encrypted per psychologist.
- Hash-chained append-only audit log and a transactional event store for replay.
- CORS lockdown, per-IP rate limiting, input sanitization (SQL-injection and XSS rejection), and global error handlers that never leak stack traces.
- Signed, expiring trusted-contact links (HMAC) that cannot be guessed, replayed, or repurposed.

Security subsystems are continuously benchmarked - PBKDF2 derivation cost, Fernet round-trip overhead, JWT handshake, and encrypted storage scaling are part of the nightly validation suite (see the Validation & Timing Report).

7. Measured Performance

Sentinel ships with a repeatable benchmark harness (backend/benchmarks) that records latency, concurrency, and resource usage in an IRIS-style logbook. Results from the latest automated run:

Subsystem	Measurement	Result
Discrepancy engine (rule-based)	50 profiles	0.1 ms, 96% accuracy
Crisis engine concurrency	1-25 concurrent	102-109 ms, 0 dropped
Storage I/O (JSON/SQLite)	10-500 profiles	23-181 ms round-trip
AI summarizer (mock baseline)	100-500 words	50-52 ms
AI summarizer (Ollama local)	100-500 words	1.4-5.1 s TTFT
PBKDF2 key derivation	10k-600k iterations	4-156 ms
Fernet crypto round-trip	100B-100KB	0.3-1.8 ms
JWT auth handshake	64B-1KB claims	0.5 ms typical

Live end-to-end measurements on the running deployment show REST endpoints at 7-41 ms, login (password derivation) at ~397 ms, and the full AI triage pipeline at ~4.7 s. Full methodology, raw run IDs, and pass/fail accounting are documented in the companion Validation & Timing Report.

8. Deployment and Total Cost of Ownership

The platform is designed for zero-cost infrastructure:

- Backend: FastAPI + SQLAlchemy over SQLite/PostgreSQL (open source).
- Frontend: React 19 + TypeScript + Vite, installable PWA with offline app shell and network-first API caching.
- AI: local `sentinel` model via Ollama (fully offline) or opt-in Groq Cloud free tier.
- Hosting: bare-metal, Docker Compose, or Render.com (free HTTPS, auto-deploy).
- Hardware: consumer smart rings via the RingSource SDK; deterministic simulator for development and CI.

The on-premises posture means clinics and research institutions can operate the full system without any subscription, per-seat, or inference-per-call cost - a material difference for low-resource settings.

9. Roadmap

Hardware M0 (ring SDK, device binding, authenticated ingestion) is complete and validated. M1-M3 extend ingestion to live BLE and vendor cloud streams, add multi-device pairing and battery telemetry, and move the adaptive sensory room (light slewing, acoustic masking, olfactory modulation) from prototype to clinical pilot. Software work targets expanded longitudinal analytics, standardized export (FHIR-adjacent), and multi-site PostgreSQL deployment.

10. Conclusion

Sentinel demonstrates that comprehensive, privacy-preserving mental health infrastructure is achievable on commodity hardware at zero recurring cost. Its differentiation is architectural: tri-directional stakeholder connectivity, unified psychophysiological signal fusion, deterministic crisis escalation, and an AI layer that assists without deciding. The full implementation - source, tests, and benchmarks - is open and auditable.

Companion documents: Research Paper (docs/sentinel_paper.pdf), Validation & Timing Report (sentinel_validation_report.pdf), Technical Design (docs/TECHNICAL_DESIGN.md). Copyright 2026 Sentinel Ecosystem (Independent Research). Licensed under the Apache License, Version 2.0.